

Peak Detection using Transformers

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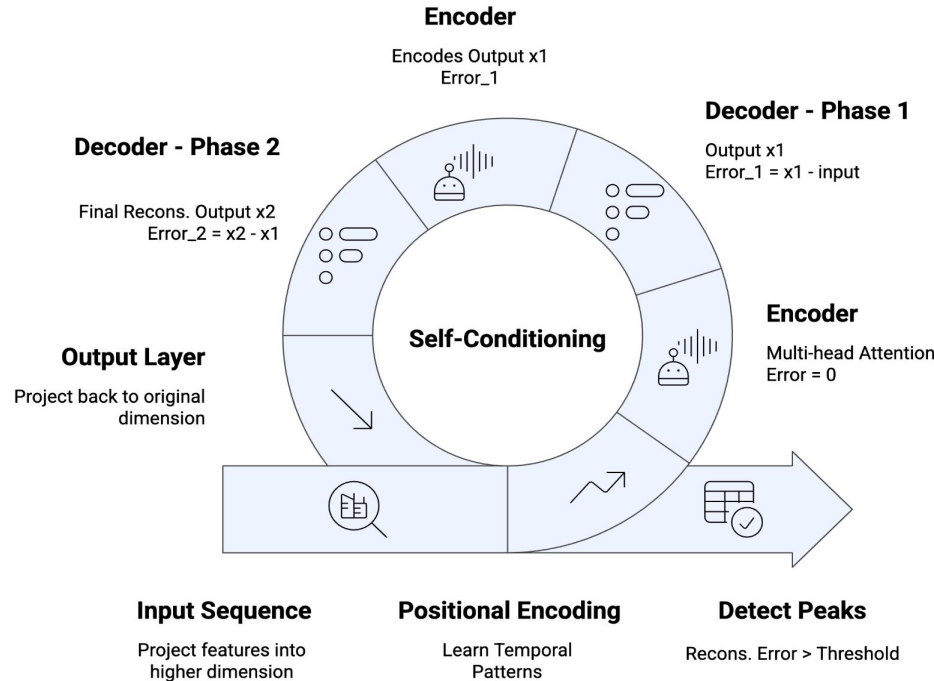
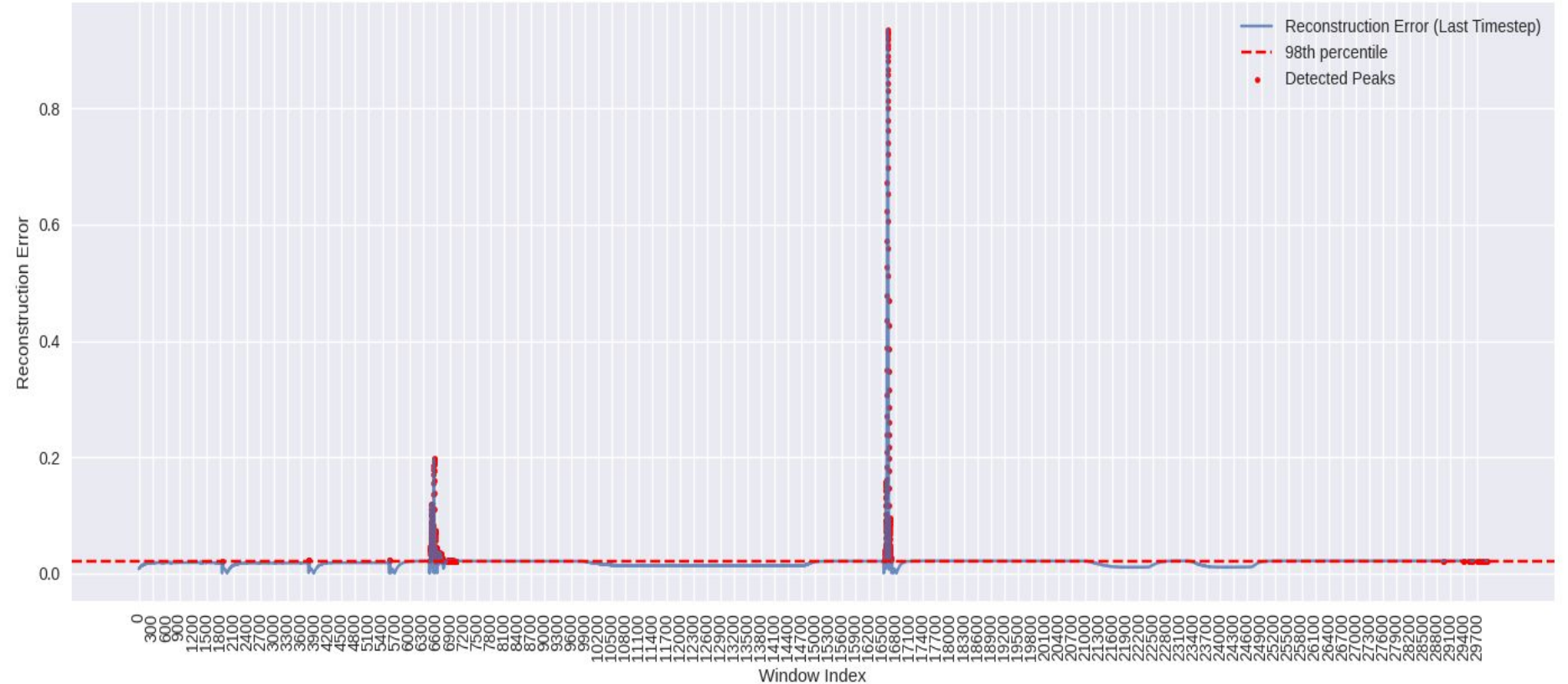


Fig : TranPD Model

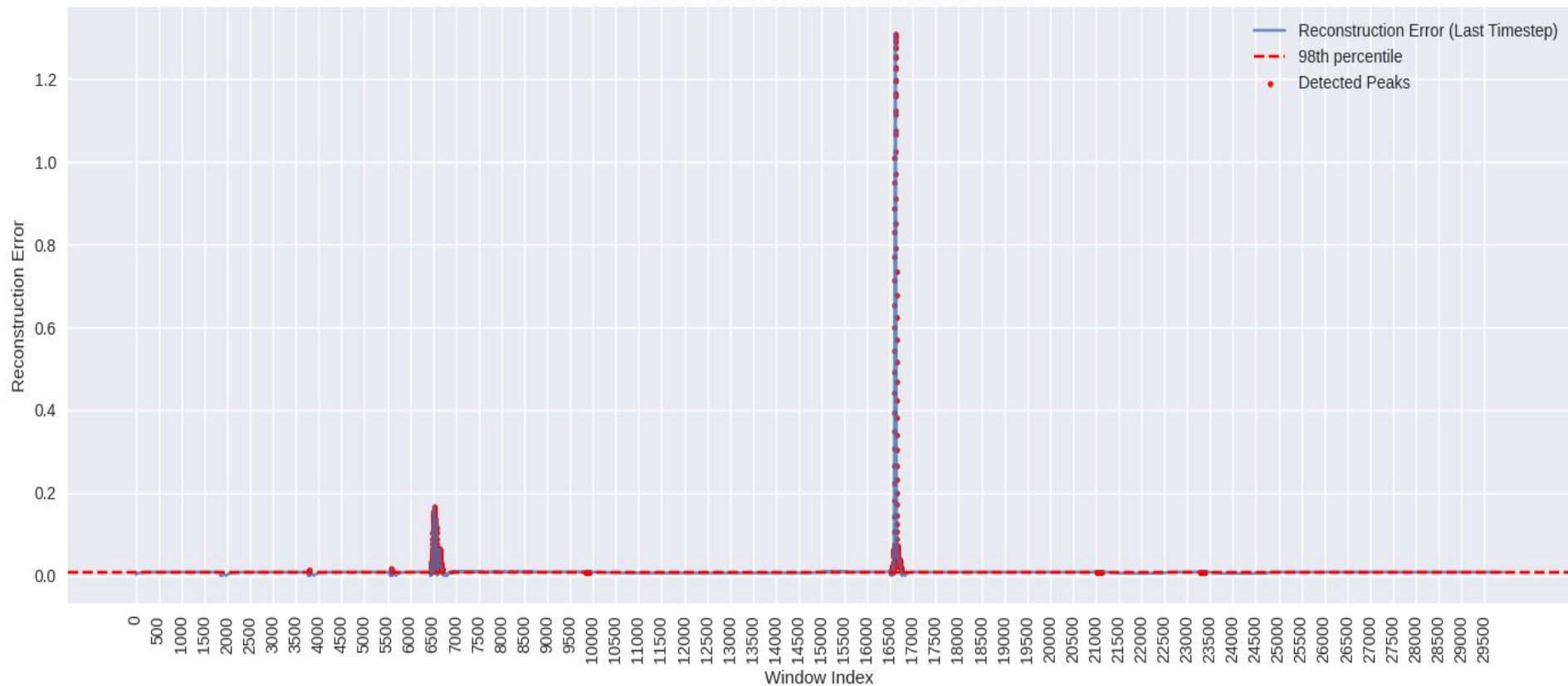
- Peak = Identifying local maxima or minima
- 2017 Paper - **Attention Is All You Need!!!**
- Powers models like BERT, GPT, ViT etc.
- NLP, Computer Vision, Time-Series Forecasting

Train Test Split	Training : Reconstruction	Random Forest with Loss	Random Forest without Loss	MAE, R ²
Chronological	W _s = 60	Num of Features = 53	Num of Features = 52	Comparing Accuracy for Both.
	Dim = 16	Target = Current 6	Target = Current 6	
Train : Test = 25173 : 16783	Trained Model with Training Set	n_estimators = 100	n_estimators = 100	
	Reconstruction on Training Set + Reconstruction on Test Set			
	RE for complete Dataset			

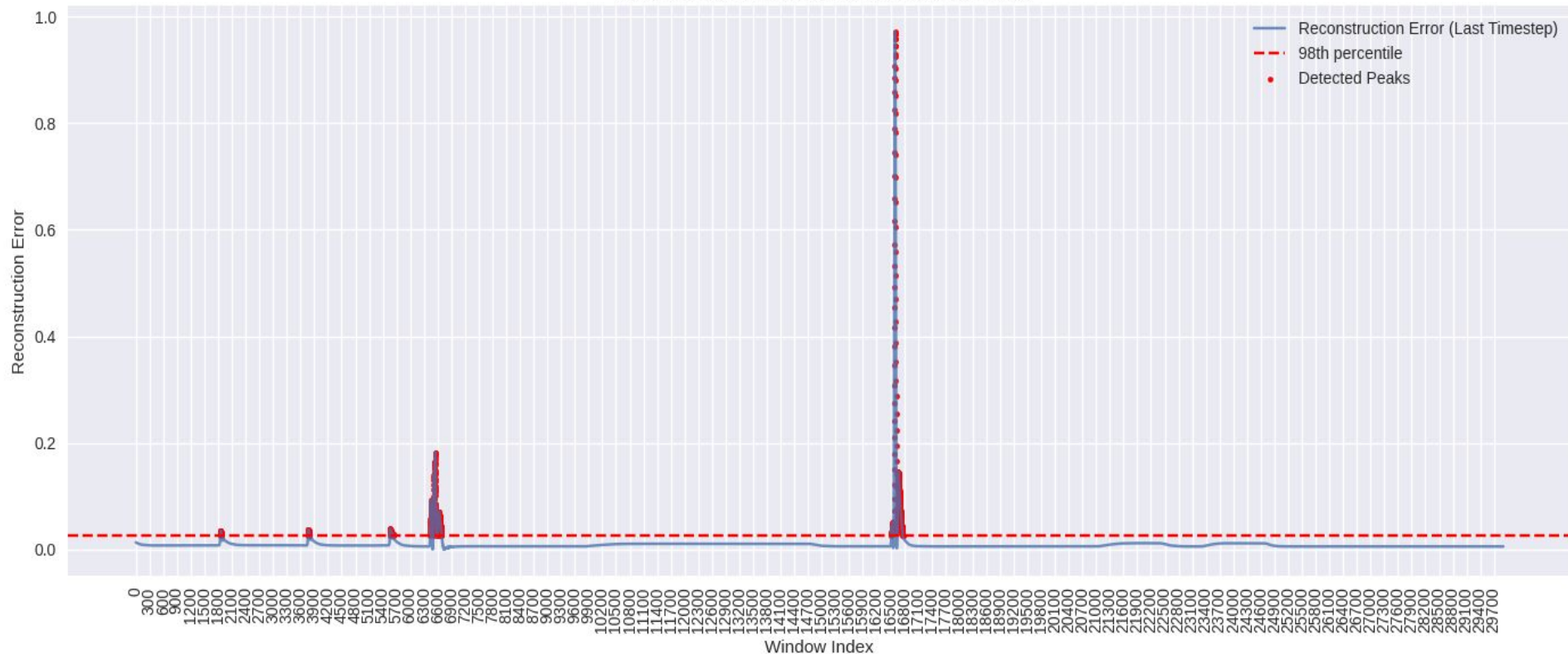
Tran-PD: Peak Detection via Reconstruction Error



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Tran-PD: Peak Detection via Reconstruction Error



	Num of Features	MAE	RMSE	R^2
RF with Loss	53	0.1128	0.8497	0.9340
RF without Loss	52	0.1244	0.8379	0.9358

Dataset	Features	MAE	RMSE	R ²
DMC2_S_CP2	52	0.1244	0.7020	.9358
DMC2_S_CP2	52 + RE	0.1128	0.8497	0.9340
DMC2_S_CP2	52 + Zscore	0.3262	0.5711	0.9702
DMC2_S_CP2	52 + RE + Zscore	0.0465	0.3437	0.9686

Results for Train:Test Split

Dataset, Size	Train:Test Split Ratio	Training Size	Test Size	W_size, Dim	No. of Peaks
DMC2_S_CP2, 42016	60: 40	25192	16794	W_size=30 Dim=16	336
DMC2_S_CP2, 42016	40:60	16794	25192	W_size=30 Dim=16	504
DMC2_S_CP2, 42016	20:80	8396	33588	W_size=30 Dim=16	672
CMX1_AL_CP2, 74822	60:40	44857	29905	W_size=30 Dim=16	599
CMX1_AL_CP2, 74822	60:40	44857	29905	W_size=60 Dim=16	599

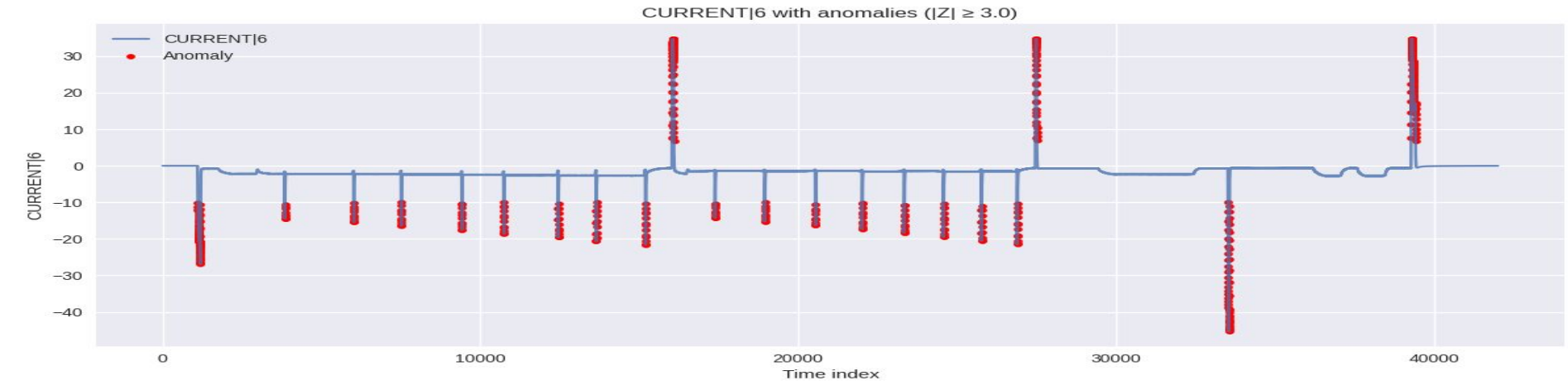
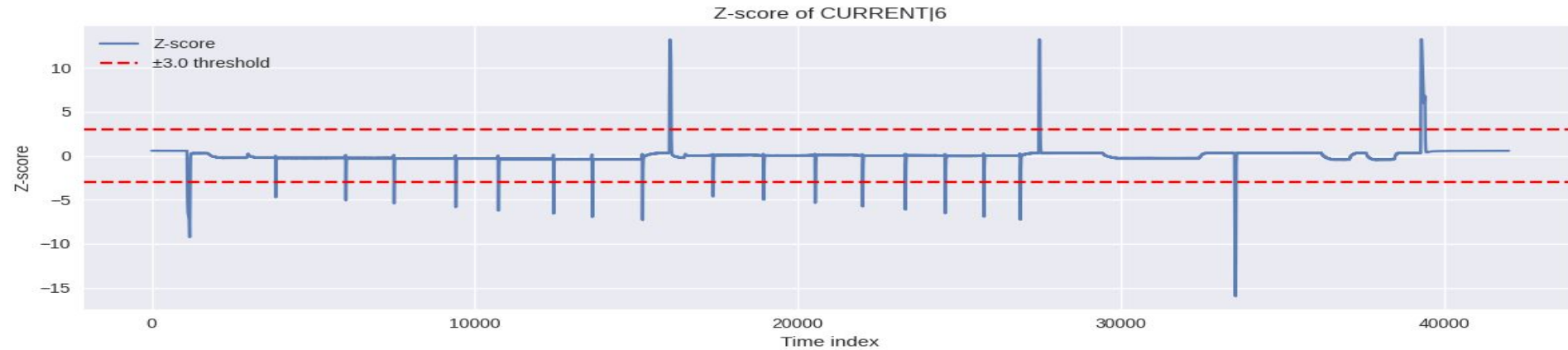


Thank You For Your Attention!

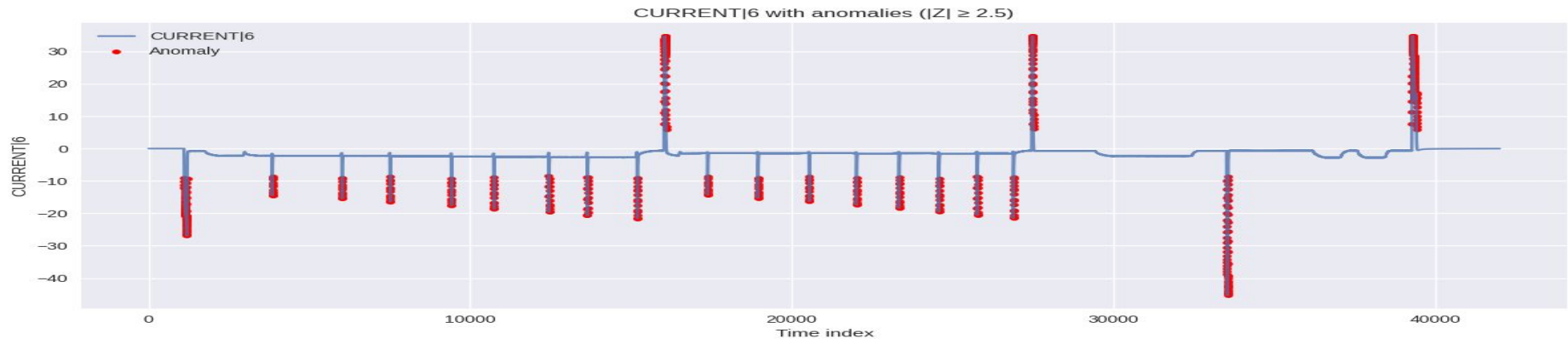
Measure	W = 120, D= 64	W = 120, D= 32	W = 120, D= 16	W = 60, D= 32	W = 60, D= 16	W = 30, D= 32	W = 30, D= 16
No. of Peaks	336	336	336	336	336	336	336
Threshold	0.82	0.11	0.08	0.23	0.15	0.17	0.06
Mean RE	0.12	0.03	0.07	0.12	0.13	0.03	0.02
Min RE	0	0.00	0	0	0	0	0 - perfect zero
Max RE	10.44	7.50	10.55	8.558	8.559	9.33	9.40

Table : Statistical Analysis of Reconstruction Error

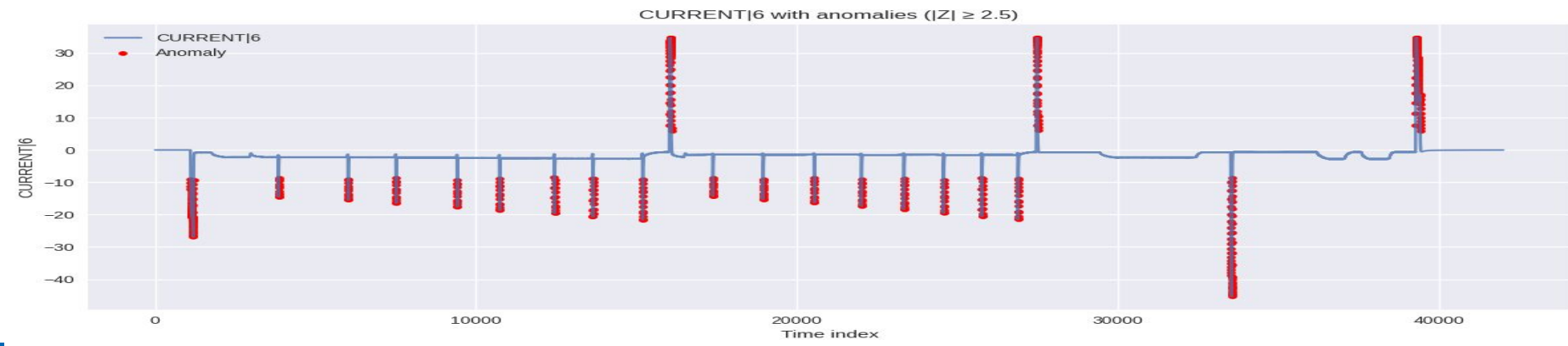
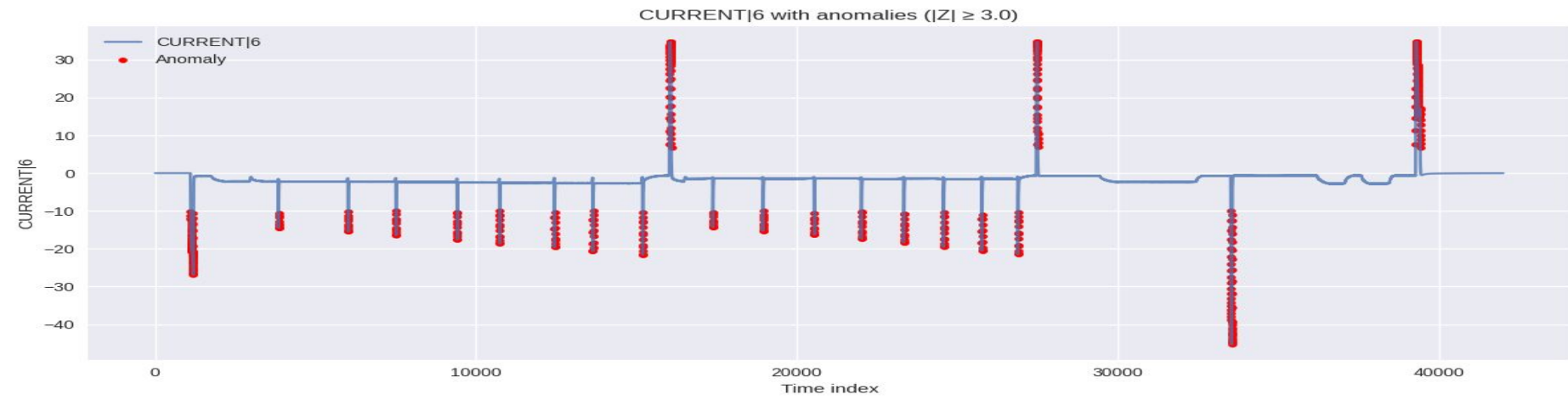
Z-Score : Threshold - 3, Num of Peaks = 548



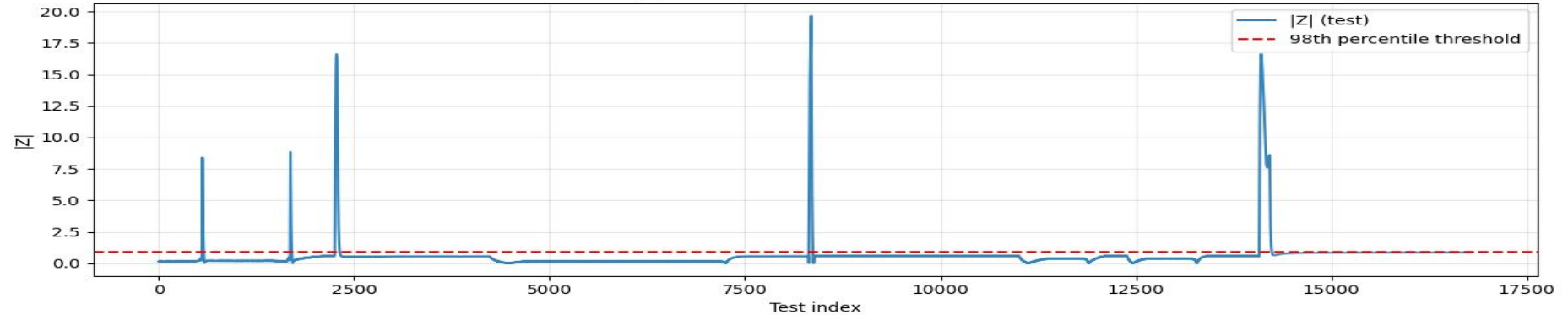
Z-Score : Threshold - 2.5, Num of Peaks = 586



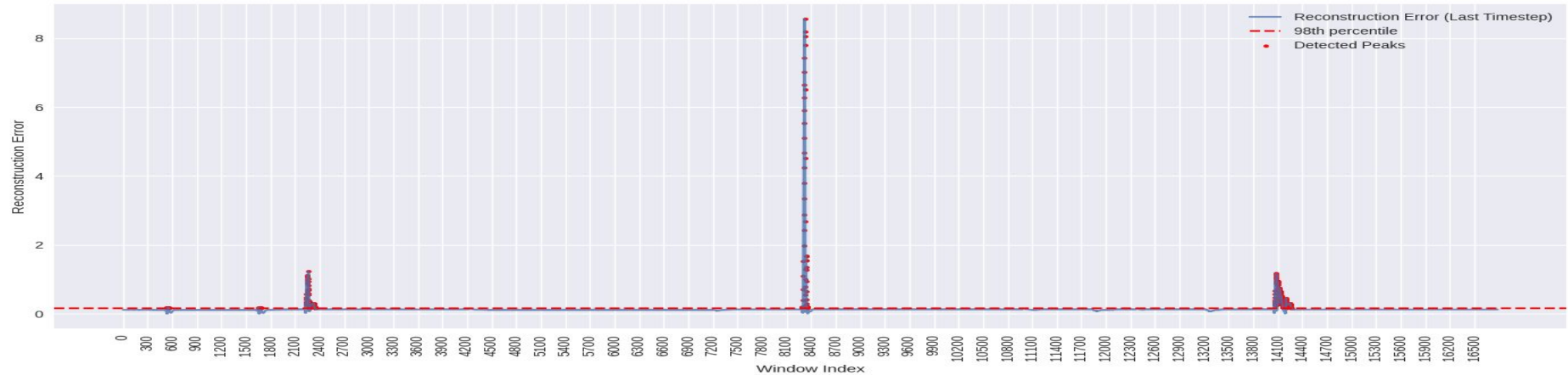
Z-Score : Threshold - 3 Vs. Threshold - 2.5



$|Z|$ Test with Percentile Threshold



Tran-PD: Peak Detection via Reconstruction Error



	Z-Score	TranformerPD
No of Peaks - Test Dataset	396	336
Mean	Z-Score : 0.6360	RE Mean - 0.13
Threshold 98%	Z-score Threshold - 0.8816	RE Threshold - 0.1565

1. Regression Analysis
2. Implement Adaptive Thresholding Mechanism
3. Multivariate Peak Detection
4. Peak Forecasting
5. regression using Loss and without Loss try to predict current and later on all cure=rent features
6. z score vs DLM - Prediction power (R^2)

1. S. Tuli, G. Casale, and N. R. Jennings, "TranAD: Deep Transformer Networks for Anomaly Detection in Multivariate Time Series Data," *Proc. VLDB Endowment (PVLDB)*, 2022. Available: <https://doi.org/10.48550/arXiv.2201.07284>
2. Z. Z. Darban, G. I. Webb, S. Pan, C. C. Aggarwal, and M. Salehi, "Deep Learning for Time Series Anomaly Detection: A Survey," *arXiv preprint arXiv:2211.05244*, 2024]. Available: <https://doi.org/10.48550/arXiv.2211.05244>
3. <https://github.com/imperial-qore/TranAD>
4. <https://www.geeksforgeeks.org/data-analysis/peak-signal-detection-in-real-time-time-series-data/>
5. <https://stackoverflow.com/questions/22583391/peak-signal-detection-in-realtime-timeseries-data>